

IVÁN SOTILLO DEL HORNO

Software Engineer | Backend & Data Systems | Applied Computer Vision

📍 Madrid, Spain (Open to relocation) | ✉ ivandelhorno@gmail.com

🌐 [linkedin.com/in/ivansotillo](https://www.linkedin.com/in/ivansotillo) | 🐙 github.com/LittleHaku | 🌐 ivansotillo.com

EXPERIENCE

Software Engineer, Computer Vision & Data

May 2026 – Present

Gambooza

Madrid, Spain

- Built and productionized a multi-tenant POS ingestion and reconciliation pipeline using Google Cloud Functions and Firebase Firestore, integrating BDP, Ágora, Square, and Glop to process ~72k–180k POS sale lines per week across 18 venues.
- Automated weekly POS-to-computer-vision discrepancy reporting, reducing report preparation from roughly half a workday to ~10 minutes while scaling coverage from 10 to 18 venues (+80%).
- Designed the POS data model and normalization layer for noisy catalogs of 100–700 products per venue, mapping 10–20 beer products/taps per venue to support dashboards and client-facing reports.
- Built POS-to-computer-vision reconciliation and beer-counting logic on top of YOLO detections and tracking IDs, achieving 87% overall counting precision, with 67% of evaluated videos reaching 94% count accuracy.

Software Engineer / Research Staff

Dec. 2024 – Apr. 2026

Universidad Autónoma de Madrid – MISO Research Group

Madrid, Spain

- Built and deployed a full-stack chatbot testing platform with React, Tailwind CSS, Django REST, PostgreSQL, Celery, Docker, and Nginx.
- Developed TRACER and LLM-based profiling pipelines with LangGraph and OpenAI APIs to explore chatbots, extract functional models, generate test scenarios, and automate releases with GitHub Actions and PyPI publishing.

PROJECTS

CertiCamper | Full-Stack Web Platform [WEBSITE]

Present

- Developing a platform for a vehicle engineering consultancy that replaces fragmented email and document workflows with a guided customer portal for camper van homologation.
- Building the system with FastAPI, React, TypeScript, PostgreSQL, Docker, Cloudflare R2, authentication, RBAC, quote flows, and dynamic forms.

Obsidian Smart Export | TypeScript [CODE] [WEBSITE]

Present

- Built and maintain an open-source Obsidian Community Plugin with nearly 3k downloads that exports structured knowledge graphs to Markdown.
- Implemented BFS-based graph traversal over wikilinks to preserve note structure, plus GitHub Actions release automation.

EDUCATION

Universidad Autónoma de Madrid

Madrid, Spain

MSc in Data Science

Sep. 2024 – Jun. 2025

- GPA: 8.41/10; thesis (10/10), published: *Automated Exploration of Conversational Agents for the Synthesis of Testing Profiles*; Highest Honors in Recommendation Systems.

Universidad Autónoma de Madrid

Madrid, Spain

BSc in Computer Science

Sep. 2020 – Jun. 2024

- GPA: 8.31/10; Highest Honors in Data Structures and Mobile App Development; Excellence Scholarships in 2nd and 4th year; international study in Finland and China.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript/TypeScript, Java, C/C++, Kotlin, SQL, Bash

Backend & Systems: Google Cloud Functions, Firebase Firestore, Django, FastAPI, REST APIs, PostgreSQL, Celery, Docker, Nginx

Integrations: TPV/POS systems, data ingestion, data normalization, scheduled jobs, API integration

Frontend: React, Vue, Astro, Tailwind CSS

AI / Data: Computer vision, YOLO, object tracking, POS-to-vision reconciliation, DINO, Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, LangGraph, OpenAI API

Tools: Git, GitHub Actions, Linux, Vim, VS Code, Jupyter, LaTeX